Science G10

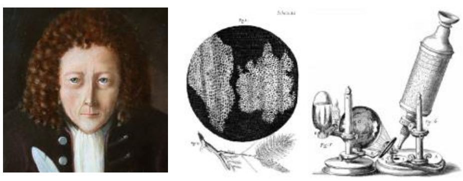
Systems of Living Things
Class 4

Agenda

1. Take up Class 2 Homework
2. Quiz 4
3. Lesson 4
   - Cell Theory
   - Animal and Plant Cells
   - Cell Division and Cell Cycle
   - Cancer
   - Stem Cells
   - Hierarchy of an Organism
   - Types of Tissues

The Cell Theory

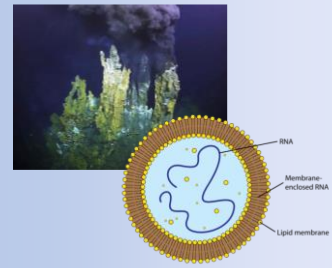
- **The Cell Theory** states that:
  - All living things are made up of one or more cells
  - The cell is the simplest unit that can carry out all life processes
  - All cells come from pre-existing cells



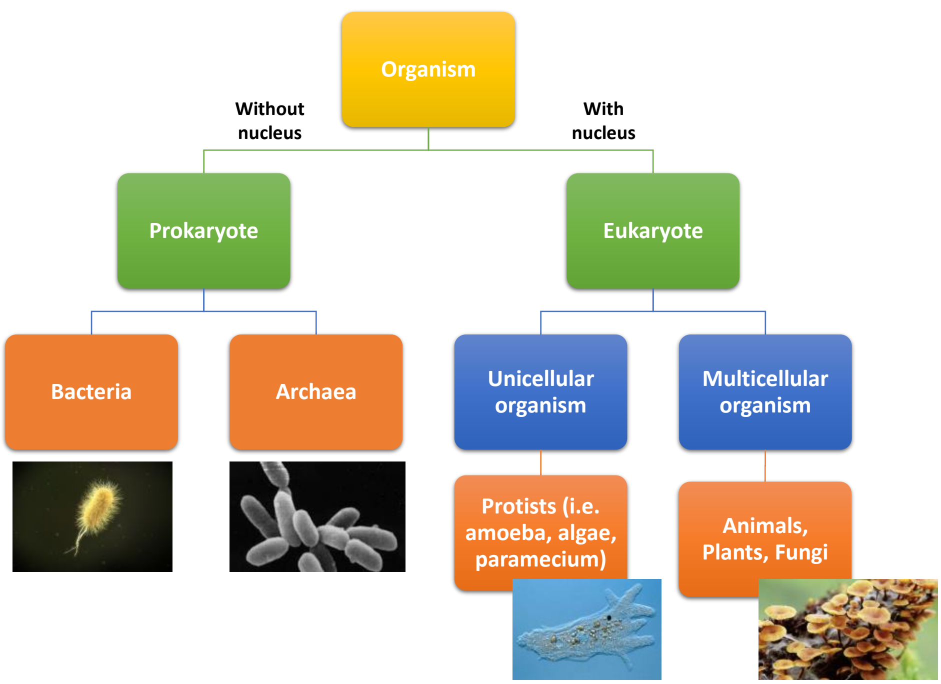
Robert Hooke (left) invented a simple microscope to look at thin slices of cork.

Classical cell theory was proposed by Schwann, Schleiden and Virchow.

If cells must come from pre-existing cells, where did the first cell come from?

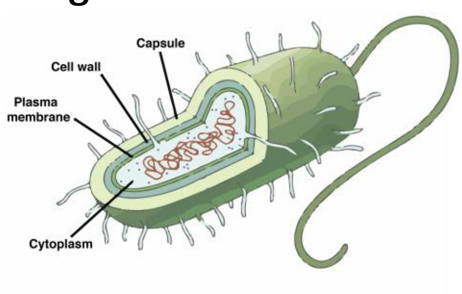


- It is proposed that life appeared 3.8 billion years ago
- Hydrothermal vents created a electrical gradient to form the first organic molecules such as amino acids and nucleotides
- First cell is presumed to be self-replicating RNA enclosed by phospholipids

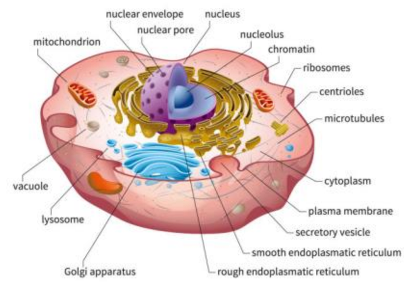


Prokaryotes and Eukaryotes

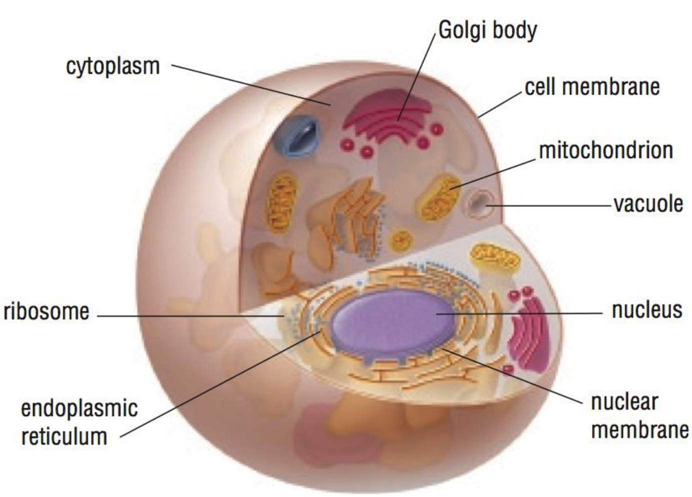
Prokaryote
- No nucleus
- No membrane-bound organelles



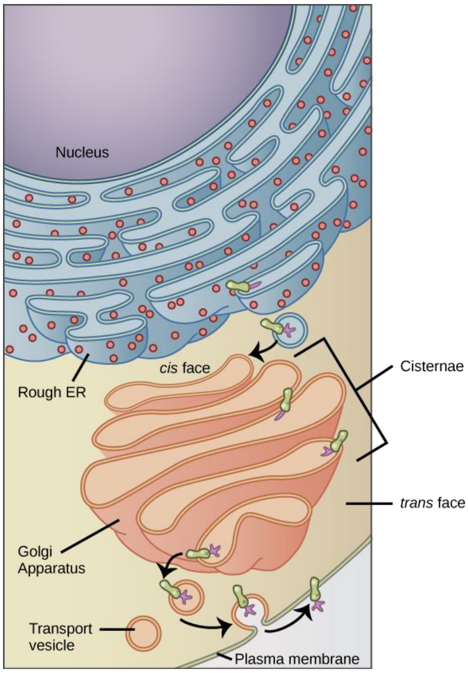
Eukaryote
- Has nucleus
- Has membrane-bounded organelles



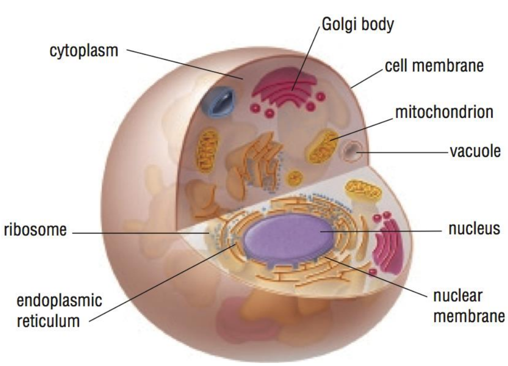
Eukaryote → Animal Cell

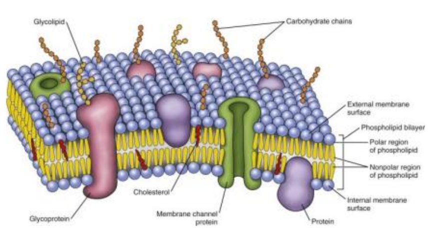


Organelles:
- **Nucleus** stores genetic material in the form of DNA
- **Nuclear membrane** controls what enters and exits the nucleus
- **Cytoplasm** is a gel-like substance that suspends organelles; provides water and nutrients



- **Endoplasmic reticulum (ER)** transports materials from the nucleus to the cell membrane
- **Ribosomes** make proteins from nucleus material
  - Ribosomes attached to ER make proteins destined for outside the cell
  - Ribosomes in the cytoplasm make proteins destined for the cell itself
- **Golgi body** collects materials from the ER and processes them for removal from the cell
- **Vesicle** are small, sacs that transport material and eventually fuses with the cell membrane

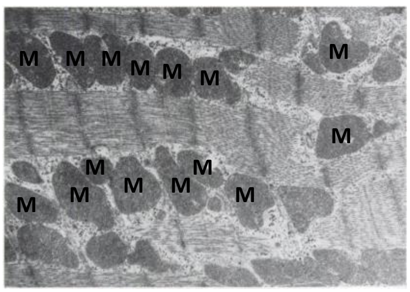


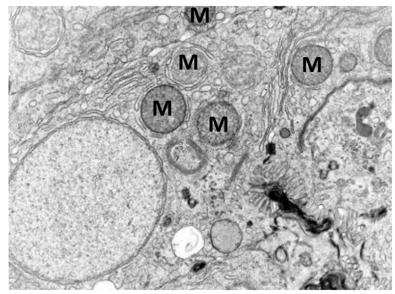


- **Vacuole** are larger sacs that store water and nutrients; does not fuse with cell membrane
- **Mitochondria** converts stored energy into usable energy through cellular respiration
- **Cell membrane** consists of a semi-permeable double membrane to control what enters and exits the cell

Checkpoint

The electron micrographs below depict two types of cells – heart muscle cell and liver cell. Which one do you think represents the heart muscle cell and why?
(M = mitochondria)

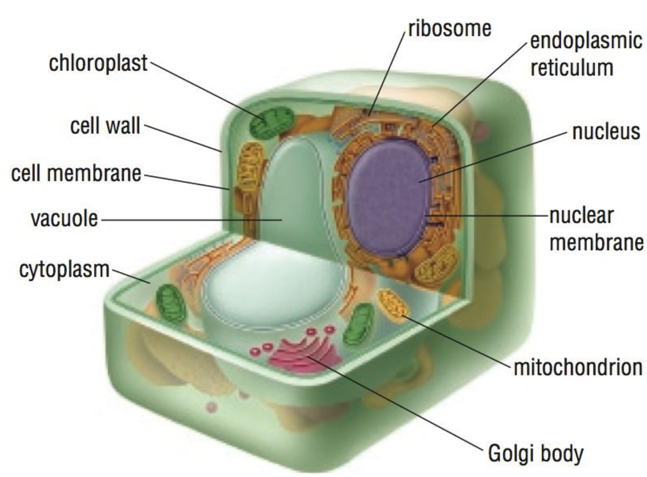




Eukaryote → Plant Cell

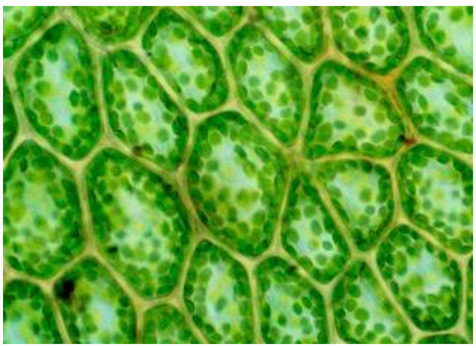
Similar organelles to animals cells with the addition of:

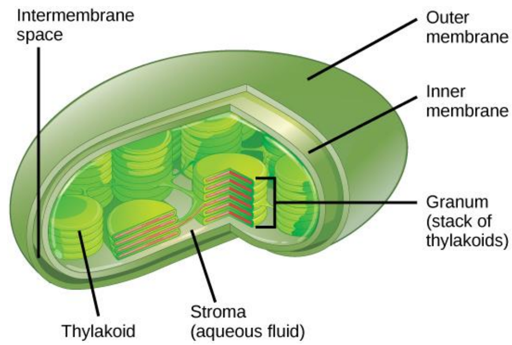
- **Cell wall** is a rigid and porous structure made of cellulose that surrounds the cell membrane
- **Large vacuole** that stores water and maintains turgor pressure



• **Chloroplast** contains pigment called chlorophyll that absorbs light energy to undergo photosynthesis

$$6CO_2 + 6H_2O + \text{light} \rightarrow C_6H_{12}O_6 + 6O_2$$





Checkpoint

If organelles worked at a factory, which organelle would be assigned the following jobs?

Factory Job	Cell Organelle
Manager's office	
Electricity	
Assembly line	
Quality control (finishing/packaging) department	
Courier (delivery)	
Kitchen	
Factory floor	

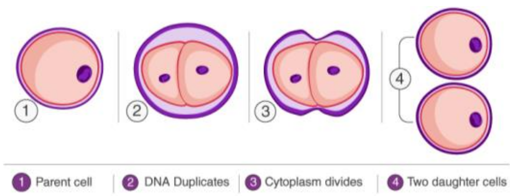
Cell Division

Cell division allows organisms to:

1) Reproduce

Asexual reproduction is the process of producing offspring from only one parent; offspring are exact genetic copies of the parent

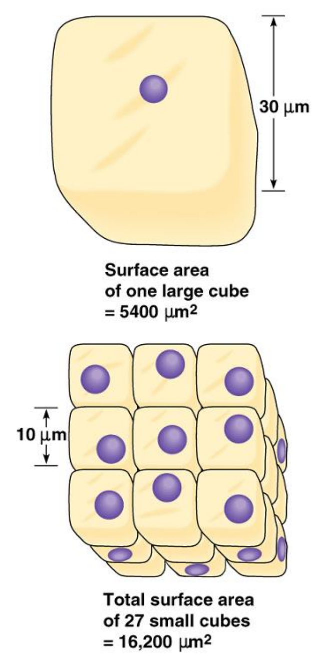
Sexual reproduction is the process of producing offspring by the fusion of two gametes (egg and sperm); offspring have genetic material from both parents



Cell division allows organisms to:

2) Grow

- Nutrients and waste move through **diffusion** (high concentration of solute to low concentration of solute) in the cell
- Water enters and leaves the cell through **osmosis** (high concentration of water to low concentration of water)
- Cells needs to divide to ensure a high surface area to maximize diffusion of nutrients/waste and osmosis of water

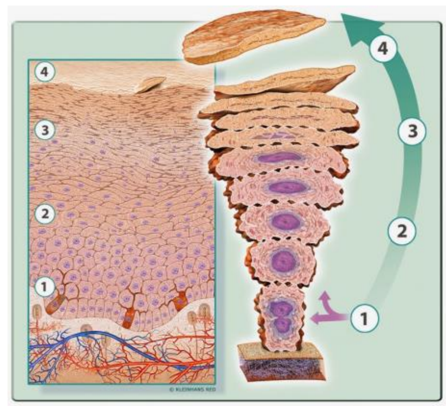


Cell division allows organisms to:

3) Repair

- Organism needs to repair cells from damage or old age

Cell Type	Turnover Time
Stomach	2-4 days
Skin	10-30 days
Red blood cells	4 months
Liver cells	6-12 months
Brain cells	Lifetime



Stages of Skin Regeneration

1. New skin cells
2. Coming to surface
3. Degeneration process
4. Dead skin cells

Cell Cycle

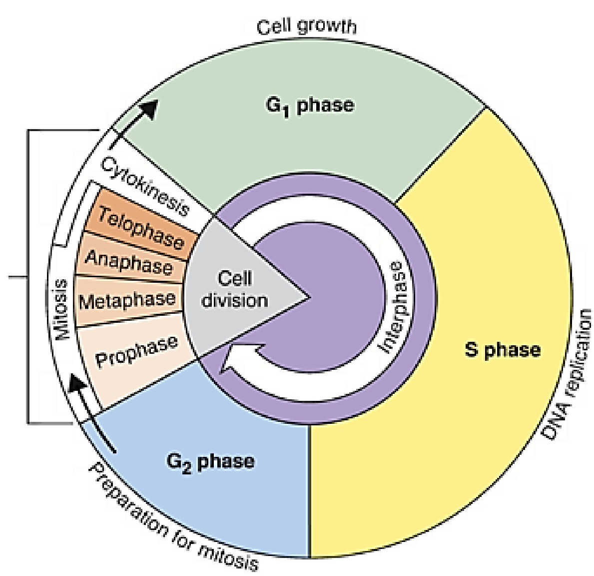
Three stages:

1) Interphase

2) Mitosis

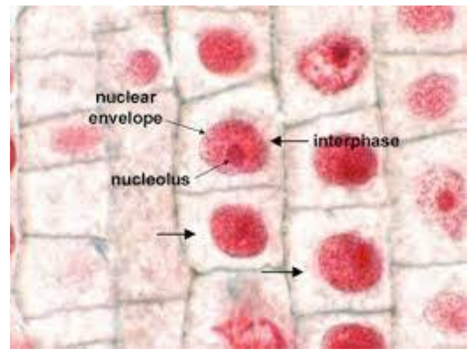
- Prophase
- Metaphase
- Anaphase
- Telophase

3) Cytokinesis



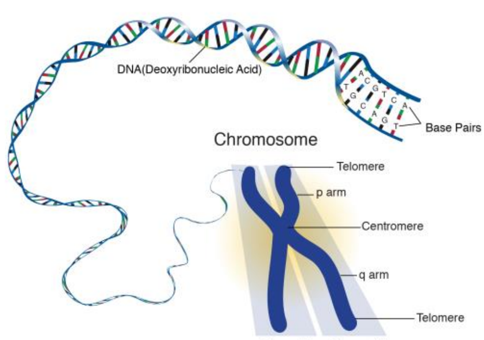
1) Interphase

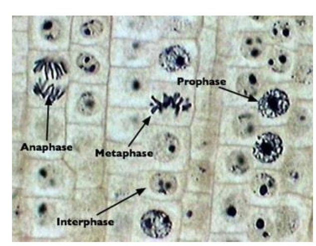
- **Interphase** is the phase in which the cell grows, performs its regular function, and copies its genetic material for cell division
  - **G_1 phase (First Gap)** is when the cell grows
  - **S phase (Synthesis of DNA)** is when the DNA replicates
  - **G_2 phase (Second Gap)** is when organelles are duplicated



2) Mitosis

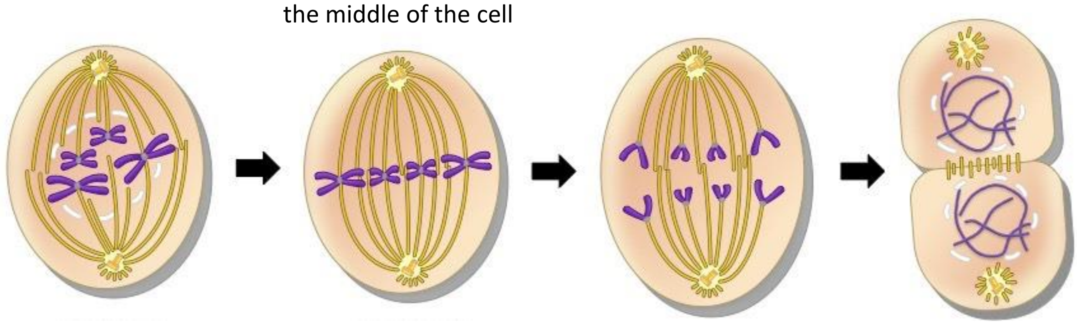
- **Mitosis** is the stage in which the DNA in the nucleus is divided
- Consists of:
  - **Prophase**
  - **Metaphase**
  - **Anaphase**
  - **Telophase**





Metaphase
- Chromosomes line up in the middle of the cell

Telophase
- Chromosomes unwind
- Nuclear membrane reforms

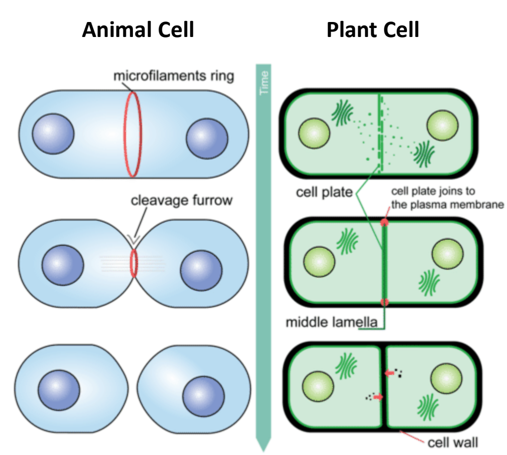


Prophase
- Nuclear membrane dissolves
- DNA became thick and visible called **chromosomes**

Anaphase
- **Centromere** splits and sister chromatids separate and move to opposite ends

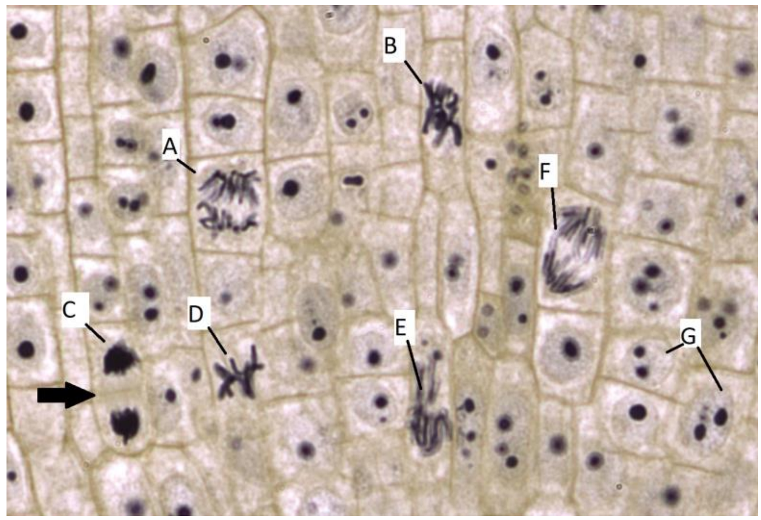
3) Cytokinesis

- **Cytokinesis** is when the cell divides producing two genetically identical daughter cells
  - In animal cells, cell membrane is pinched off in the centre
  - In plant cells, a division plate grows from the centre towards the cell walls



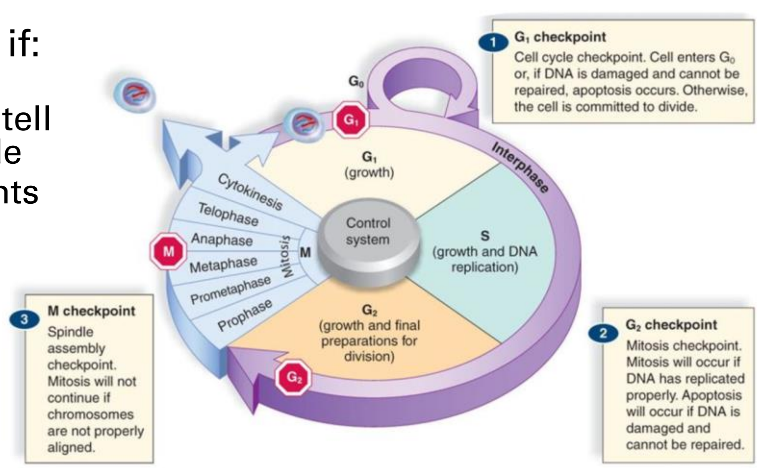
Checkpoint

Which stage of the cell cycle does each letter represent?



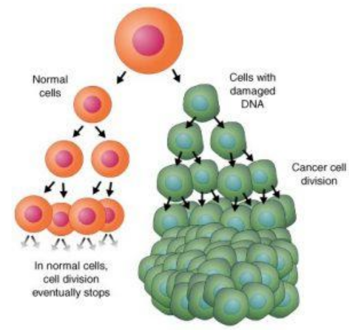
Healthy Cell Checkpoints

- A cell will not divide if:
  - Signals from surroundings cells tell the cell not to divide
  - Not enough nutrients for cell growth
  - DNA has not been replicated
  - DNA is damaged
- Cell can undergo programmed cell death (apoptosis)



Cancer

- **Cancer** is a group of diseases in which cell grows and divides uncontrollably
- Caused by mutations to the DNA within cells which can allow rapid growth, fail to stop uncontrolled cell growth, or make errors when correcting DNA
- Mutations are inherited or caused by environmental factors (i.e. radiation, viruses, carcinogens, etc.)



Stages of Cancer

Stage 0

Cancer cells remain in place to form a mass of cells called a **tumour**

Stage I

Small tumour has not grown deeply into nearby tissues and has not spread to lymph nodes

Stage II and III

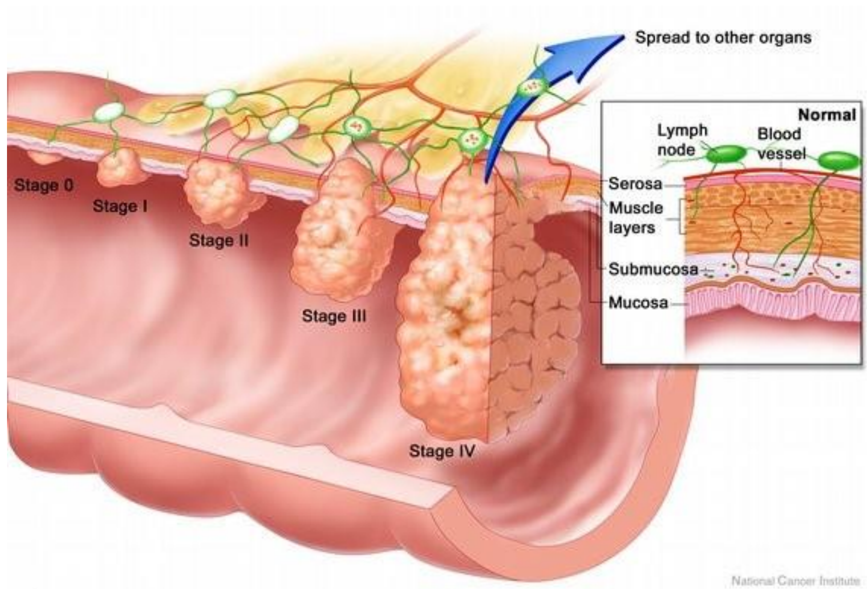
Larger tumours have grown deeply into nearby tissue and may have spread to lymph nodes

Stage IV

Cancer has spread to other organs or parts of the body **(metastasis)**

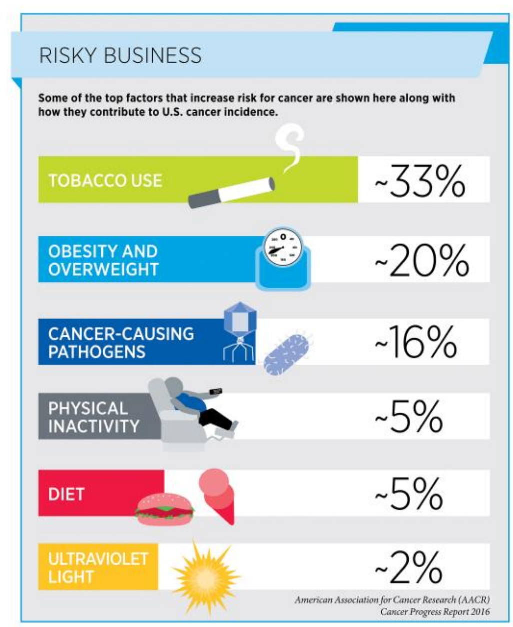
Colorectal Cancer 5-year Survival Rates

Stage	Survival Rate
I	94%
II	82%
III	67%
IV	11%



Cancer Prevention

- Healthy choices:
  - Live smoke-free
  - Wear sunscreen and sun protection
  - Maintain a healthy body weight
  - Get vaccinated
  - Check your family history
  - Get screened regularly

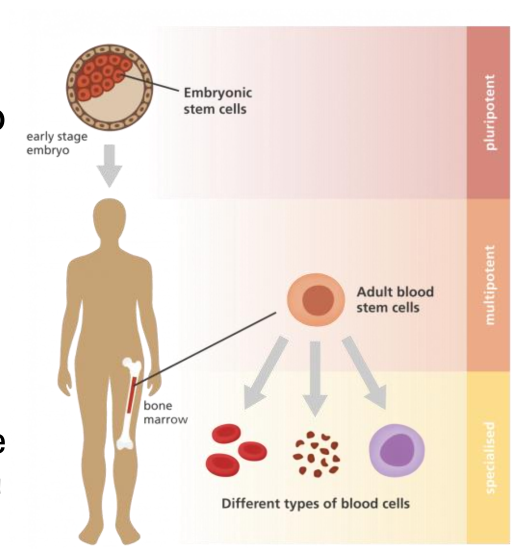


Cancer Treatment

- **Surgery** to physically remove the tumour
- **Chemotherapy** uses chemicals and drugs to kill cancer cells; taken orally or intravenously
- **Radiation therapy** uses focused beams of radiation to target cancer cells

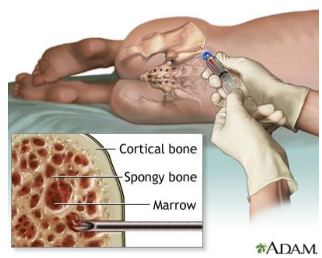
Stem Cells

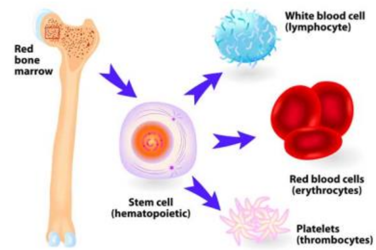
- **Stem cells** are cells that can self-renew and differentiate into many different types of cells in the body
- Two types:
  - **Embryonic stem cells** are derived from embryos and can differentiate into any cells
  - **Adult stem cells** are found in the body and can differentiate into a limited number of cell types



Stem Cell Transplant

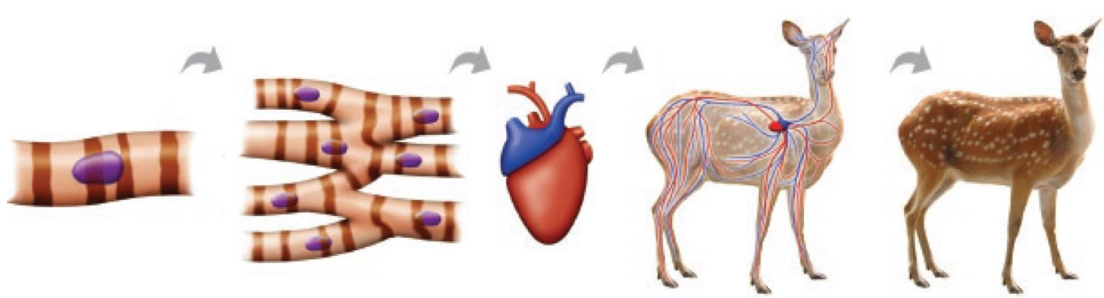
- **Stem cell transplant** replaces stems cells from a donor and is performed when a patient's stem cells or bone marrow have been damaged by disease, chemotherapy, or radiation therapy
- Stem cells can be collected from the:
  - Bone marrow
  - Peripheral blood
  - Umbilical cord





Hierarchy of an Organism

Cell → Tissue → Organ → System → Organism



(a) Cellular level heart muscle cell

(b) Tissue level heart muscle tissue

(c) Organ level heart

(d) Organ system level circulatory system

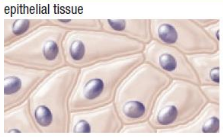
(e) Organism level deer

- **Cell –** the smallest functioning unit of an organism
- **Tissue –** a group of cells that perform a similar, limited function
  - Four main types: Epithelial, Connective, Muscle, and Nerve tissue
- **Organ** – a structure composed of different tissues to perform a complex function
- **Organ System** – a system of one or more organs that work together to perform a vital bodily function

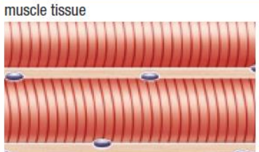
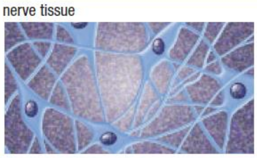
Tissues

- **Epithelial tissue** is a thin sheet that covers body surfaces and lines internal organs
- **Connective tissue** provides support and protection

Table 1 Animal Tissue Types

Type	Example	Description	Function
epithelial tissue 	• skin • lining of the digestive system	• thin sheets of tightly packed cells covering surfaces and lining internal organs	• protection from dehydration • low-friction surfaces
connective tissue 	• bone • tendons • blood	• various types of cells and fibres held together by a liquid, a solid, or a gel, known as a matrix	• support • insulation

- **Muscle tissue** contains proteins like actin and myosin that can contract and move
- **Nerve tissue** conducts electrical signals from one part of the body to another

muscle tissue 	• muscles that make bones move • muscles surrounding the digestive tract • heart	• bundles of long cells called muscle fibres that contain specialized proteins capable of shortening or contracting	• movement
nerve tissue 	• brain • nerves in sensory organs	• long, thin cells with fine branches at the ends capable of conducting electrical impulses	• sensory • communication within the body • coordination of body functions

Checkpoint

Why is blood considered connective tissue?

What I Learned Today:

- ❑ Cell Theory
- ❑ Animals and Plant Cells
- ❑ Reasons for Cell Division
- ❑ Cell Cycle
- ❑ Cancer
- ❑ Stem Cells
- ❑ Types of Tissues

Due next class: Class 4 Homework